

Design Note: Caching Strategy for the Read Path

We evaluated three caching approaches for the product catalog read path: cache-aside, write-through, and write-behind. Write-through was selected because it keeps the cache consistent with the database on every write, which matters for pricing data. The Redis cluster runs three nodes with replication enabled. Invalidation uses a 15-minute TTL as a safety net. Estimated infrastructure cost is 220 USD per month in the staging tier.